Block practice · Further body systems

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Name receptors and the senses or stimuli they detect

Compare blood entering and leaving the kidney

Interpret a raised or lowered blood cell count

Explain stereoscopic vision and how it helps judge distance

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Practice workbook · Name receptors and the senses or stimuli they detect

Naming the structure that detects a stimulus, listing senses, naming skin temperature receptors, and reasoning about having several cone types.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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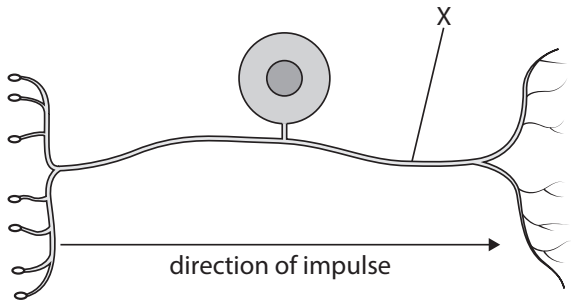
Type ID: HBI-NAME-RECEPTOR-AND-SENSE

1. Question 1 · part c.iii · 1 mark

June 2024 Paper 1

Worked-example retry · Attempt part c.iii only.

1 (a) The diagram shows a cell found in the human nervous system.



(Source: © Designua/ Shutterstock)

(i) The cell shown in the diagram transmits electrical impulses from a receptor cell to the central nervous system.

State the name of this type of nerve cell.

(1)

.....

(ii) Name the structure labelled X.

(1)

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1. Question 1 · part c.iii · 1 mark

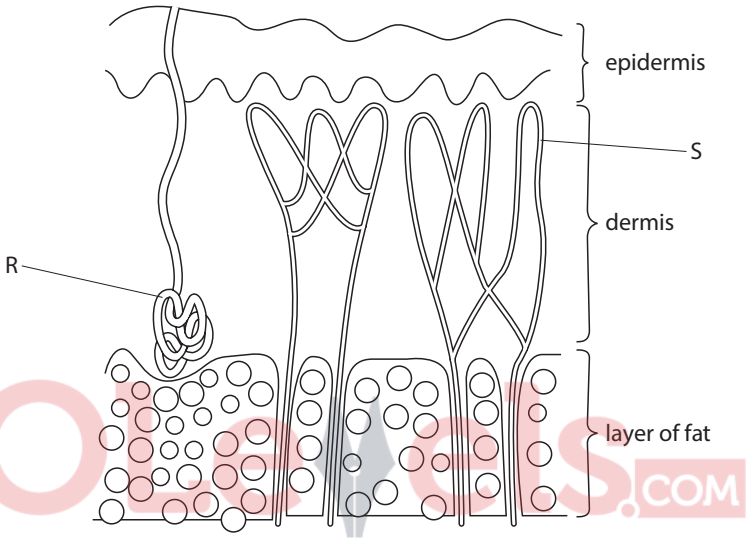
June 2024 Paper 1

Worked-example retry · Attempt part c.iii only.

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(b) Structures in the skin respond to information from the nervous system.

The diagram shows a section of the skin.



(Source: adapted from Human Physiology and Health, Author David Wright, Publisher Heinemann page 113)

(i) Name part R and part S.

(2)

part R

.....

part S

.....

(ii) Describe how part R helps to cool the body.

(2)

.....

.....

.....

.....

1. Question 1 · part c.iii · 1 mark

June 2024 Paper 1

Worked-example retry · Attempt part c.iii only.

(iii) Describe the role of the layer of fat in maintaining body temperature.

(2)

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1. Question 1 · part c.iii · 1 mark

June 2024 Paper 1

Worked-example retry · Attempt part c.iii only.

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(c) A student investigates the effect of temperature on receptors in the skin.

This is their method.

- place one hand in hot water
- place the other hand in ice cold water
- after 2 minutes take both hands out of the water
- place both hands in water at room temperature

(i) State one possible hazard with their method.

(1)

.....

.....

(ii) Describe how to reduce the risk of this hazard.

(2)

.....

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.....

(iii) When the student placed both hands in the water at room temperature the water felt hot to one hand and ice cold to the other.

Name the structure in the skin that detects temperature.

(1)

.....

.....

(Total for Question 1 = 12 marks)

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1. Question 1 · part c.iii · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 4 · part a · 2 marks

June 2023 Paper 1

Further practice · Attempt part a only.

4 The eye contains receptors that can detect light.

(a) Name two other senses.

(2)

1 .....

2 .....

(b) The table shows how the focal length of the eye lens changes as the thickness of the lens changes.

Thickness of lens in mm	Focal length in mm
10	12.5
9	14.0
8	15.5
7	18.0
6	21.0
5	25.0

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(i) Describe the relationship between the thickness of the lens and its focal length.

(2)

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(ii) Give the effect on the diameter of the lens as its thickness increases.

(1)

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2. Question 4 · part a · 2 marks · Working and answer

Working and answer

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3. Question 7 · part a.i · 1 mark

June 2024 Paper 1R

Further practice · Attempt part a.i only.

7 (a) The nervous system sends electrical impulses to parts of the body in response to an external stimulus.

(i) Name the structure in the nervous system that detects an external stimulus.

(1)

.....

(ii) Name the structure that transmits electrical impulses directly to the central nervous system.

(1)

.....

(b) Neurones are specialised cells.

(i) Describe what is meant by a specialised cell.

(2)

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3. Question 7 · part a.i · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 4 · part c.iii · 2 marks

June 2023 Paper 1

Further practice · Attempt part c.iii only.

4 The eye contains receptors that can detect light.

(a) Name two other senses.

(2)

1 .....

2 .....

(b) The table shows how the focal length of the eye lens changes as the thickness of the lens changes.

Thickness of lens in mm	Focal length in mm
10	12.5
9	14.0
8	15.5
7	18.0
6	21.0
5	25.0

OLevels.COM SCHOOL

(i) Describe the relationship between the thickness of the lens and its focal length.

(2)

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(ii) Give the effect on the diameter of the lens as its thickness increases.

(1)

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4. Question 4 · part c.iii · 2 marks

June 2023 Paper 1

Further practice · Attempt part c.iii only.

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(iii) A person is looking at a distant object and then looks at a near object.

Use the data in the table to explain the changes that occur in the eye so the person is able to form an image of the near object on the retina.

(4)

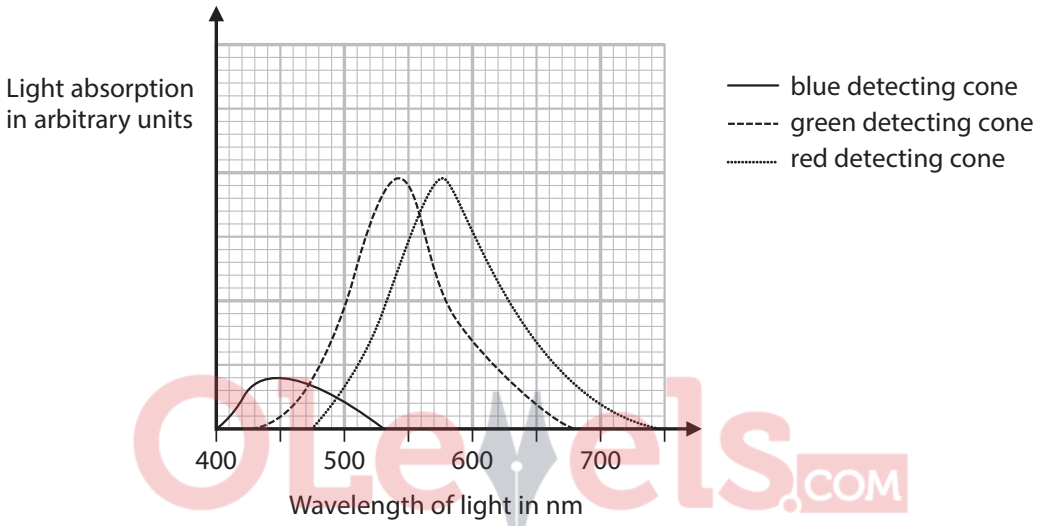
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4. Question 4 · part c.iii · 2 marks

June 2023 Paper 1

Further practice · Attempt part c.iii only.

(c) Cone cells are found in the retina of the eye. They detect colour. The graph shows the absorption of different wavelengths of light by three types of cone cell.



(i) Determine the wavelength of light at which a red detecting cone absorbs most light.

(1)

wavelength of light = ... nm

(ii) Determine the maximum difference in the wavelengths of the light that can be absorbed by blue detecting cones.

(2)

difference in wavelengths = ... nm

(iii) Suggest why it is better to have three types of cone rather than just one type of cone.

(2)

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(Total for Question 4 = 14 marks)

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4. Question 4 · part c.iii · 2 marks · Working and answer

Working and answer

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Practice workbook · Compare blood entering and leaving the kidney

Differences between renal artery and renal vein blood, and what urine solutes are dissolved in.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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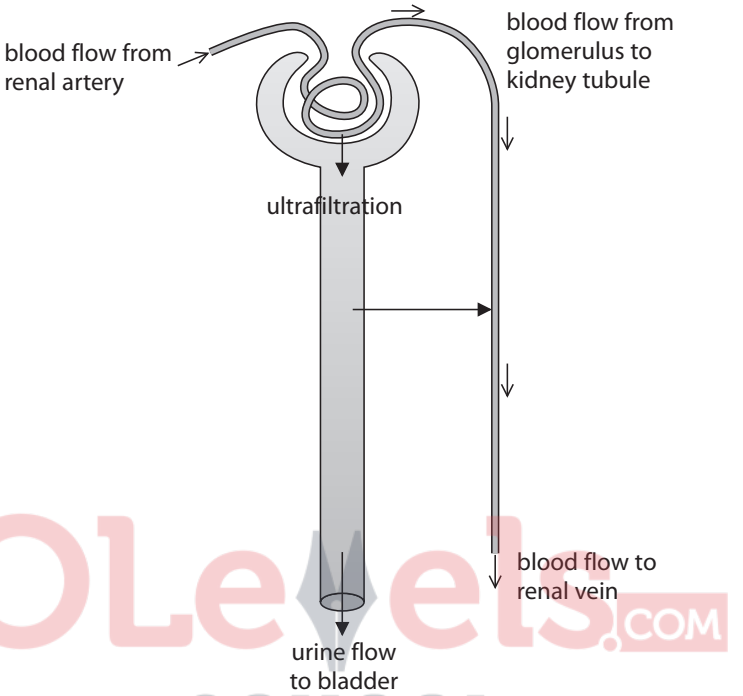
Type ID: HBI-COMPARE-RENAL-BLOOD-COMPOSITION

1. Question 8 · part a.ii · 4 marks

June 2019 Paper 2

Worked-example retry · Attempt part a.ii only.

8 (a) The diagram shows part of a nephron in the human excretory system.



(Source: adapted from © Alila Medical Images/Alamy)

(i) Explain how ultrafiltration occurs in the Bowman's capsule.

(2)

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1. Question 8 · part a.ii · 4 marks

June 2019 Paper 2

Worked-example retry · Attempt part a.ii only.

(ii) Explain the differences in the composition of the blood flowing through the renal artery and through the renal vein.

(4)

(b) The quantity of salt in the diet affects the concentration of urine produced.

An investigation is set up to test this statement.

This is the method used.

- ten people are each given a different food
- each food contains a different quantity of salt
- the volume of urine produced by each person is recorded each hour for six hours after eating the food

(i) State the independent variable in this investigation.

(1)

(ii) Explain why this investigation is unlikely to produce valid results.

(2)

(iii) State one change to the method that would make the results more valid.

(1)

(Total for Question 8 = 10 marks)

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1. Question 8 · part a.ii · 4 marks · Working and answer

Working and answer

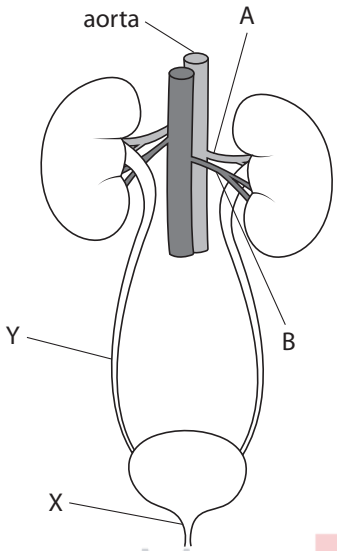
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2. Question 8 · part a.iii · 2 marks

January 2020 Paper 1

Further practice · Attempt part a.iii only.

8 The diagram shows part of the human renal system.



(a) (i) Give the names of structures X and Y. **(2)**

structure X .....

structure Y .....

(ii) Describe the function of the aorta. **(2)**

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(iii) Compare the composition of the blood in vessels A and B. **(2)**

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2. Question 8 · part a.iii · 2 marks

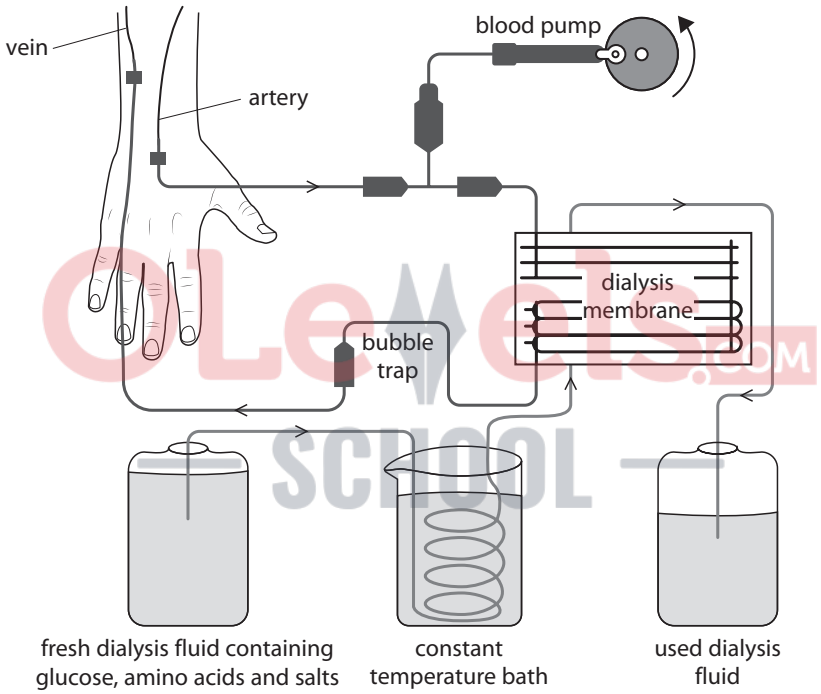
January 2020 Paper 1

Further practice · Attempt part a.iii only.

(b) Dialysis is a treatment used for people with kidney failure.

Blood containing waste substances is pumped through a dialysis machine. The dialysis machine filters the waste from the blood. The cleaned blood is then returned to the patient.

The diagram shows how a dialysis machine filters the blood of a person with kidney failure.



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2. Question 8 · part a.iii · 2 marks · Working and answer

Working and answer

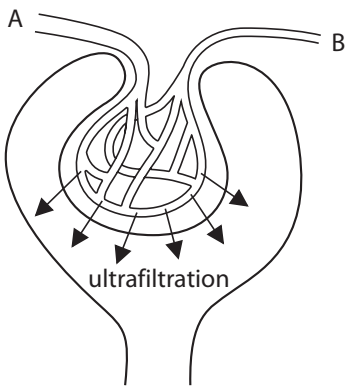
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3. Question 6 · part a.ii · 3 marks

June 2021 Paper 1

Further practice · Attempt part a.ii only.

6 (a) The diagram shows a Bowman's capsule and its associated blood supply.



(i) Describe what is meant by the term **ultrafiltration**.

(2)

(ii) State three differences between the composition of blood in vessel A and the composition of blood in vessel B.

(3)

1 .....

.....

2 .....

.....

3 .....

.....

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3. Question 6 · part a.ii · 3 marks

June 2021 Paper 1

Further practice · Attempt part a.ii only.

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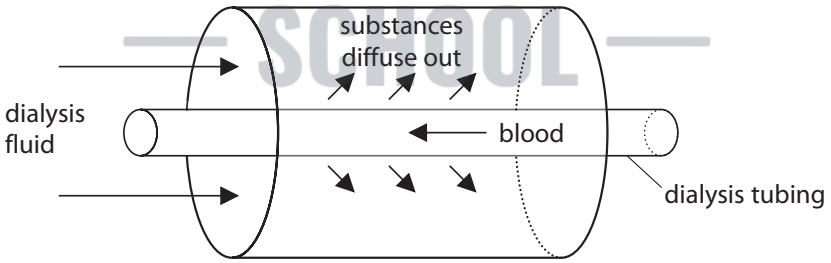
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(iii) Explain how blood vessel A and blood vessel B help the process of ultrafiltration.

(4)

(b) The diagram shows part of a kidney dialysis machine.



The dialysis fluid contains glucose at the same concentration as in the blood flowing through the dialysis tubing.

Suggest the effect on a person undergoing dialysis if the concentration of glucose was below that of the blood flowing through the dialysis tubing.

(3)

(Total for Question 6 = 12 marks)

3. Question 6 · part a.ii · 3 marks · Working and answer

Working and answer

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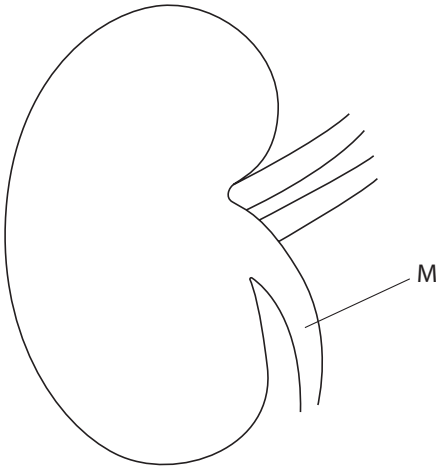
4. Question 7 · part b · 3 marks

June 2022 Paper 2R

Further practice · Attempt part b only.

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7 The diagram shows a kidney.



(a) (i) Which of these is the name of structure M?

(1)

- ☒ **A** renal artery
- ☒ **B** renal vein
- ☒ **C** ureter
- ☒ **D** urethra

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(ii) Which of these structures is connected to structure M?

(1)

- ☒ **A** bladder
- ☒ **B** prostate gland
- ☒ **C** seminal vesicle
- ☒ **D** urethra

4. Question 7 · part b · 3 marks

June 2022 Paper 2R

Further practice · Attempt part b only.

(b) State three differences between the blood entering the kidney and the blood leaving the kidney.

(3)

1 .....

.....

2 .....

.....

3 .....

.....

(c) The table compares the nitrogen content of urine during 24 hours for a person on a high protein diet with a person on a low protein diet.

Substance excreted during 24 hours	High protein diet Mass in g	Low protein diet Mass in g
total nitrogen compounds	16.8	3.6
urea	14.7	2.2
uric acid	0.18	0.09
ammonia	0.49	0.04
creatinine	0.58	0.60

(i) Calculate the percentage difference in the mass of nitrogen compounds excreted during 24 hours for the person on a high protein diet compared with the low protein diet.

(3)

percentage difference = ... %

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4. Question 7 · part b · 3 marks · Working and answer

Working and answer

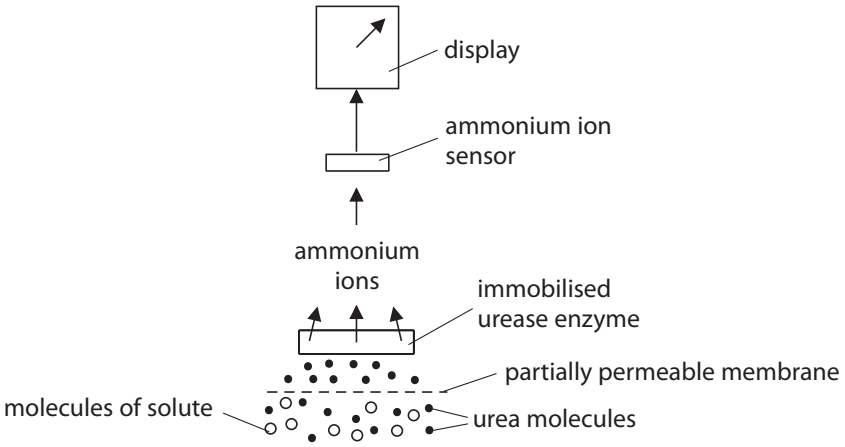
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5. Question 3 · part a.i · 1 mark

June 2021 Paper 2

Further practice · Attempt part a.i only.

3 The diagram shows a biosensor that can be used to measure the concentration of urea in urine.



(a) (i) The urea molecules are in solution.

State the name of the solvent. **(1)**

(ii) Explain the function of the partially-permeable membrane. **(2)**

(iii) Suggest the function of the urease enzyme. **(3)**

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5. Question 3 · part a.i · 1 mark · Working and answer

Working and answer

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Practice workbook · Interpret a raised or lowered blood cell count

Blood test results or ratios where white cell or platelet numbers differ from normal.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-INTERPRET-ABNORMAL-BLOOD-CELL-COUNTS

1. Question 5 · part a.ii · 2 marks

November 2024 Paper 1

Worked-example retry · Attempt part a.ii only.

5 (a) The table gives information about some blood components in three people.

Blood component	Healthy person number per cm^3	Person A number per cm^3	Person B number per cm^3
white blood cells	7500	13 750	7250
red blood cells	5.0×10^6	5.0×10^6	5.0×10^6
platelets	3.0×10^5	3.0×10^5	0.8×10^5

(i) Calculate the ratio of the number of white blood cells in the healthy person compared with the number of white blood cells in person A.
Write your answer in the form of 1 : n

(2)

1 : .....

(ii) Explain a possible reason for the difference in the number of white blood cells in the healthy person and person A.

(2)

.....

.....

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.....

(iii) Describe the effect that the lower number of platelets per cm^3 would have on the health of person B.

(2)

.....

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1. Question 5 · part a.ii · 2 marks · Working and answer

Working and answer

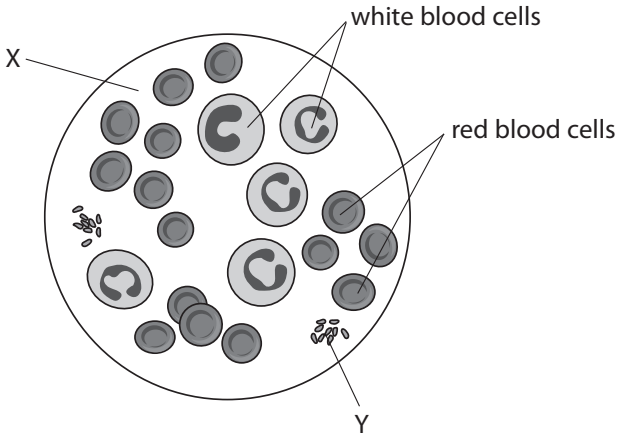
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2. Question 1 · part c.ii · 2 marks

November 2020 Paper 1R

Further practice · Attempt part c.ii only.

1 The diagram shows a sample of human blood.



(a) State the function of red blood cells.

(1)

.....

.....

(b) Give the name of X and Y.

(2)

X .....

Y .....

(c) (i) Calculate the ratio of red blood cells to white blood cells.

Write the ratio in its simplest form.

(2)

ratio = ...

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2. Question 1 · part c.ii · 2 marks

November 2020 Paper 1R

Further practice · Attempt part c.ii only.

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(ii) Suggest why this ratio will change if a person has a bacterial infection.

(2)

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(Total for Question 1 = 7 marks)

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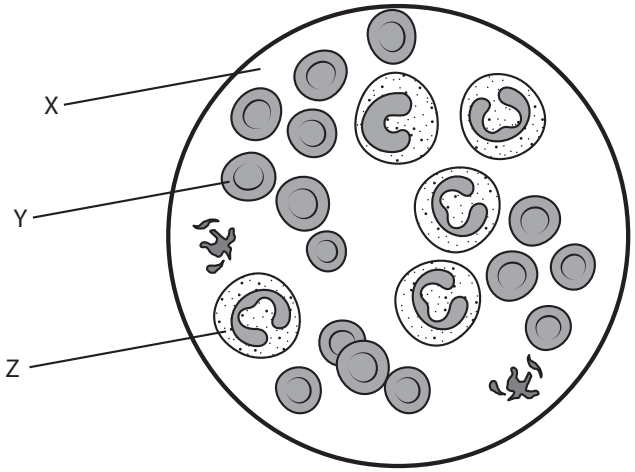
Working and answer

3. Question 2 · part a.iii · 1 mark

June 2024 Paper 1R

Further practice · Attempt part a.iii only.

2 (a) The diagram shows components, X, Y and Z, in a sample of human blood.



(i) Name components X, Y and Z.

(3)

X .....

Y .....

Z .....

(ii) Describe the function of component Y.

(2)

.....

.....

.....

.....

(iii) A sample of blood from a patient is analysed in a hospital.

A scientist finds that the number of component Z in the blood sample is higher than normal.

State a possible reason for the increase in the number of component Z.

(1)

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.....

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P 7 4 4 9 2 A 0 5 2 4

5

Turn over ▶

3. Question 2 · part a.iii · 1 mark

June 2024 Paper 1R

Further practice · Attempt part a.iii only.

(b) Blood is transported around the body inside blood vessels.

Draw **one** straight line from each blood vessel to a feature of the blood vessel.

(2)

Blood vessel	Feature of blood vessel
artery	contains valves to prevent backflow of blood
vein	has muscular walls to maintain high blood pressure
capillary	has thin walls for exchange of materials

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3. Question 2 · part a.iii · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 5 · part a.iii · 2 marks

November 2024 Paper 1

Further practice · Attempt part a.iii only.

5 (a) The table gives information about some blood components in three people.

Blood component	Healthy person number per cm^3	Person A number per cm^3	Person B number per cm^3
white blood cells	7500	13 750	7250
red blood cells	5.0×10^6	5.0×10^6	5.0×10^6
platelets	3.0×10^5	3.0×10^5	0.8×10^5

(i) Calculate the ratio of the number of white blood cells in the healthy person compared with the number of white blood cells in person A.
Write your answer in the form of 1 : n

(2)

1 : .....

(ii) Explain a possible reason for the difference in the number of white blood cells in the healthy person and person A.

(2)

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.....

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(iii) Describe the effect that the lower number of platelets per cm^3 would have on the health of person B.

(2)

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4. Question 5 · part a.iii · 2 marks

November 2024 Paper 1

Further practice · Attempt part a.iii only.

(b) (i) Haemolytic anaemia (HA) is an inherited condition.
People with this condition have a low number of red blood cells in their blood.

Explain one symptom that people with HA will have.

(4)

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4. Question 5 · part a.iii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Explain stereoscopic vision and how it helps judge distance

What stereoscopic vision is, why two eyes judge distance better than one, and an investigation comparing catching with one and both eyes.

3 question parts: 1 worked-example retry and 2 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-EXPLAIN-STEREOSCOPIC-VISION

1. Question 1 · part b · 3 marks

November 2023 Paper 2

Worked-example retry · Attempt part b only.

1 (a) The box lists parts of the eye and parts of the ear.

auditory nerve ciliary body cochlea cornea eardrum optic nerve ossicles pupil retina semicircular canals

The table has descriptions of parts of the eye and ear.

Complete the table by matching the descriptions, using parts from the box.

(7)

Description	Part
contains light sensitive cells	
vibrates when receives sound waves	
helps the body balance	
controls the amount of light entering the eye	
transfers vibrations across the middle ear	
turns vibrations into nerve impulses	
refracts light entering the eye	

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1. Question 1 · part b · 3 marks

November 2023 Paper 2

Worked-example retry · Attempt part b only.

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(b) Explain why a person using both eyes finds it easier to catch a ball than a person who is blind in one eye.

(3)

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(Total for Question 1 = 10 marks)

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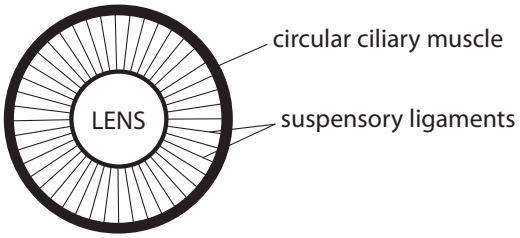
Working and answer

2. Question 5 · part b · 3 marks

January 2022 Paper 2

Further practice · Attempt part b only.

5 (a) The diagram shows a front view of structures in the eye that help to form an image on the retina.



The three structures shown in the diagram change when focusing on near and distant objects.

Complete the table by putting ticks (✓) in the correct boxes to show the condition of each structure of the eye when focusing on near and distant objects.

(3)

	Focusing on a near object	Focusing on a distant object
Lens	thin ☐ thick ☐	thin ☐ thick ☐
Circular ciliary muscle	contracted ☐ relaxed ☐	contracted ☐ relaxed ☐
Suspensory ligament	stretched ☐ loose ☐	stretched ☐ loose ☐

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P 6 7 0 6 1 A 0 1 4 2 4

2. Question 5 · part b · 3 marks

January 2022 Paper 2

Further practice · Attempt part b only.

(b) Humans have stereoscopic vision.

Explain what is meant by the term **stereoscopic vision**.

(3)

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2. Question 5 · part b · 3 marks

January 2022 Paper 2

Further practice · Attempt part b only.

(c) Design an investigation to show that it is easier for a person to catch a ball if they have stereoscopic vision.

Your answer should include experimental details and be written in complete sentences.

(6)

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(Total for Question 5 = 12 marks)

2. Question 5 · part b · 3 marks · Working and answer

Working and answer

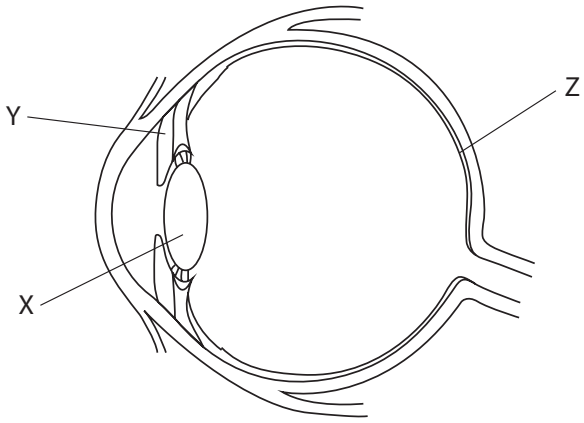
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3. Question 1 · part a.ii · 1 mark

January 2023 Paper 1

Further practice · Attempt part a.ii only.

1 (a) The diagram shows a human eye.



(i) Draw one line from each structure to its correct function.

(3)

Structure	Function
	transmits electrical impulses to the brain
structure X	refracts light rays
	protects the eye from pathogens
structure Y	
	detects light rays
structure Z	controls the shape of the lens
	controls the size of the pupil

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3. Question 1 · part a.ii · 1 mark

January 2023 Paper 1

Further practice · Attempt part a.ii only.

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(ii) Most humans have stereoscopic vision.

Which of these statements occurs because of stereoscopic vision?

(1)

- ☒ **A** being able to judge depth
- ☒ **B** being able to see in colour
- ☒ **C** having a wide field of view
- ☒ **D** to be able to see near and far objects

(b) The diagram shows a healthy eye and an eye with a cataract.



Healthy eye Eye with cataract

(i) Describe the cause of cataracts.

(2)

.....

.....

.....

(ii) State one treatment for cataracts.

(1)

.....

(iii) It is estimated that 39 million people in the world are blind.

Of these people, 43% are blind because of cataracts.

Determine the estimated number of people who are blind because of cataracts.

(2)

number of people = ...

(iv) Some people are blind because of a vitamin deficiency.

Name the vitamin that is needed to prevent this type of blindness.

(1)

.....

(Total for Question 1 = 10 marks)

3. Question 1 · part a.ii · 1 mark · Working and answer

Working and answer

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